

AMD Smart Router

Hybrid Token-Efficient Routing Agent

AMD Developer Hackathon ACT II - Track 1 - Team Verso

100.0% accuracy - 8,262 Fireworks tokens

scored on the hidden evaluation set

Docker Hub: `caverraaa/routing-agent:v4-hybrid`

GitHub: `github.com/caverraaa/AMD-Smart-Router`

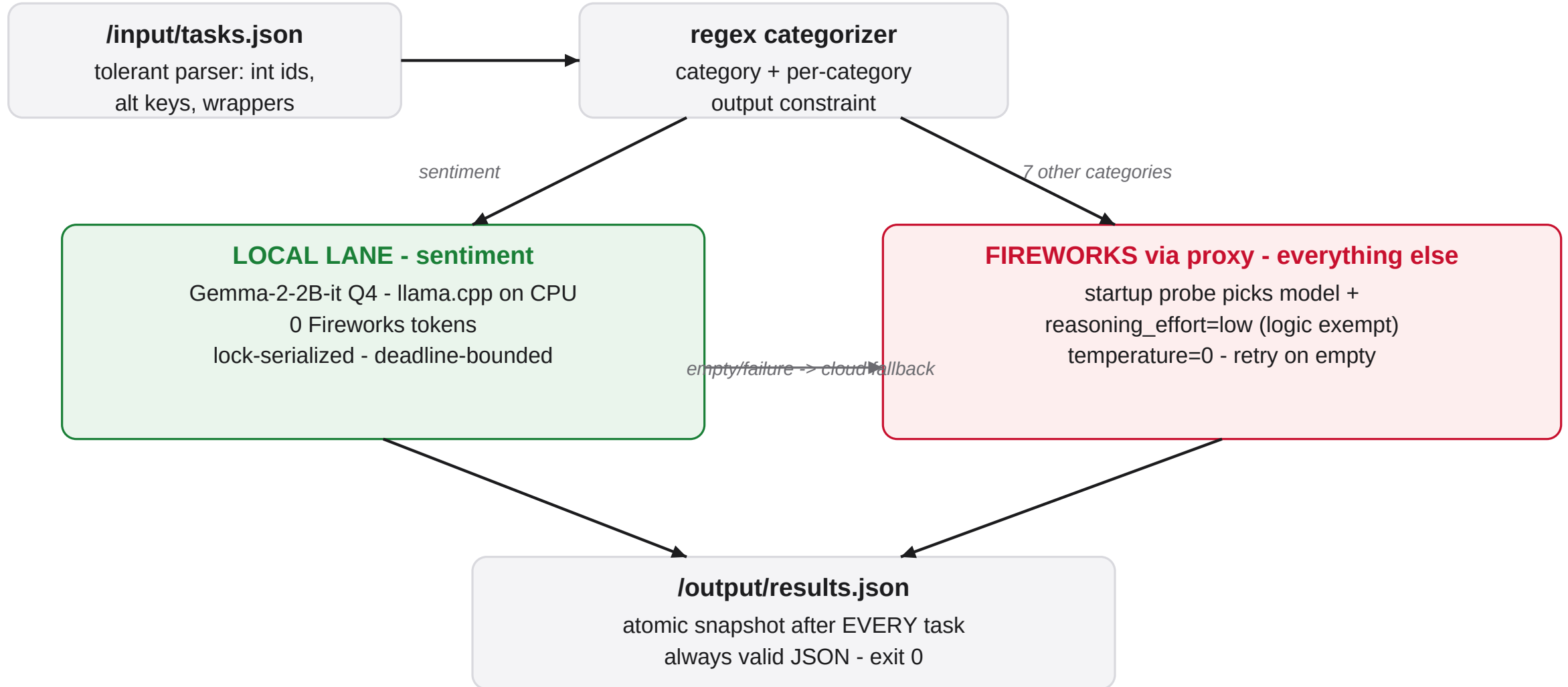
The task, and where we landed

Answer 8 categories of NL tasks with the fewest Fireworks tokens that still clear the accuracy gate - local inference is free, wrong answers are fatal.

Our submission: **100.0% accuracy at 8,262 tokens**

- We sit in the top accuracy tier of the leaderboard
- 26% fewer tokens than our own accuracy-first configuration (11,196)
- Every optimization was measured before it shipped - and one that failed our measurement bar (v5) was reverted within a single submission cycle.
- Engineering log, specs and judged experiments: all public in the repo

Architecture: one router, two lanes, always-valid output



Where the tokens went (measured, not guessed)

Hidden-reasoning probe

-~3,000

measures billed-vs-visible tokens per model at startup; applies reasoning_effort=low when accepted - reasoning was ~40% of the naive bill

Token diet

-~500

44-char system prompt + per-category output constraints - which also raised accuracy 94.7% -> 100%

Local Gemma lane

sentiment = 0 tokens

Gemma-2-2B serves the local lane, admitted per-category by a count-based judge gate: sentiment earned it 12/12; logic denied at 0/12

temperature=0

determinism

stable answers and stable measurements - score jitter across runs eliminated

Every routing decision is data: a 72-task golden set + binary-verdict LLM judge gates every change before the submission pointer moves.

Results & what we proved along the way

Version	Tokens	Accuracy	What changed
v2 baseline	8,421	94.7%	cheapest allowed model, hardened I/O
v3	11,196	100%	bought accuracy with output constraints
v4 (submitted)	8,262	100%	reasoning-tax cut + judged Gemma lane

- **Hidden eval = 19 tasks; the pass threshold is unpublished (our leaderboard-based estimate: ~80%).** Accuracy above the gate earns nothing - so we converted margin into token cuts only when the measured saving beat our noise floor.
- **Local-model frontier, measured in 4GB / 2 vCPU:**
under 2B params: not accurate enough beyond sentiment (Qwen3-1.7B: ner 7/12, math 3/6)
3B and up: too slow for long answers (Gemma-3-4B: 7-9.5 tok/s, breaks 30s/response)
- **Full engineering log - specs, judged experiments, post-mortems - public in the repo.**